**AIStudio**

# HOWTO — AIStudio User Guide

Practical answers for day-to-day AIStudio use. Not a getting-started guide (see QUICKSTART.md) — reach for this when you need to do something specific or something isn't working as expected.

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## Contents



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## Shell & Terminal

***What is the shell?*** The shell (Terminal) is a text-based interface for running commands on your Mac. AIStudio setup and management uses the shell for tasks the UI doesn't cover — like installing models or ingesting a new corpus. New to the shell? See Apple's Terminal User Guide.

***How do I open the Terminal?*** Press `Cmd+Space`, type `Terminal`, press Enter. Or open Finder → Applications → Utilities → Terminal.

***Why does my terminal say "command not found" when I paste commands?*** Two common causes: - You pasted a `#` comment line — zsh tries to execute it. Paste only the commands, not the comment lines. - Your AIStudio aliases aren't loaded yet. Run:

```
source ~/.zshrc
```

Then try the command again on its own line. Do not chain: `source ~/.zshrc && ais_start` will fail. Run `source ~/.zshrc` first, then `ais_start`.

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## Using AIStudio

After running `./ais_install` from the repo root (see QUICKSTART.md), these commands are available from any terminal. If a command is not found, run `source ~/.zshrc` first.

| Command | What it does |
|---|---|
| `ais_install [cmd]` | Install AIStudio user commands — `ais_install ais_log` adds a single command, `ais_install --verify` checks all aliases |
| `ais_start` | Start all services and open the UI in your browser |
| `ais_stop` | Stop all services |
| `ais_bench` | Run a benchmark on the demo corpus |
| `ais_log` | Tail live AIStudio backend log — run in a separate tab after ais_start |
| `ais_download_sec_10k` | Download SEC 10-K filings from EDGAR to ~/Downloads/sec_10k/ (~2 GB) |
| `ais_ingest_sec_10k` | Ingest SEC 10-K corpus into AIStudio (~30 min, backend must be running) |
| `ais_help` | Print this command reference |

Every command supports `--help`:

```
ais_start --help
ais_bench --help
```

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## Upgrading AIStudio

### How do I upgrade AIStudio when a new version is released?

Pull the latest code from GitHub:

```
cd ~/Developer/AIStudio && git pull
```

Then update Python dependencies in case new packages were added:

```
source .venv/bin/activate && pip install -r requirements.txt
```

Then restart:

```
ais_start
```

**Does upgrading delete my corpus data?** No. Your corpus data lives in `~/qdrant_storage/` on disk — completely separate from the AIStudio codebase. It persists across upgrades, restarts, and reboots. You never need to re-ingest after a routine upgrade.

**When would I need to re-ingest after an upgrade?** Only if the release notes say the chunk format changed. This is rare and always announced. When it happens, run:

```
AISTUDIO_VECTORSTORE=qdrant PYTHONPATH=src python3 -m local_llm_bot.app.ingest --corpus <
```

The `--force` flag wipes and rebuilds the corpus cleanly.

**The demo corpus re-ingests automatically** on first `ais_start` after an upgrade if needed.

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## Corpus Management

***How do I ingest a new corpus?*** Use the UI — open AIStudio, create a new corpus using the **New** button, then upload your files using the **Add** button. AIStudio handles ingestion automatically and shows progress in the chat area.

***How do I re-ingest a corpus from scratch?*** Delete the corpus using the **Delete Corpus** button in the UI (type YES to confirm — it moves to Trash, recoverable). Then create a new corpus with the same name and re-upload your files via **Add**. AIStudio ingests everything in one pass.

***What happens when I delete a file from a corpus — is it gone forever?*** No. Deleted files move to `data/corpora/<name>/uploads/trash/` — not permanently deleted.

***How do I recover a file I accidentally deleted from a corpus?***

```
# See what's in trash
ls ~/Developer/AIStudio/data/corpora/<name>/uploads/trash/

# Move it back
mv ~/Developer/AIStudio/data/corpora/<name>/uploads/trash/<filename> \
   ~/Developer/AIStudio/data/corpora/<name>/uploads/

# Re-ingest to restore it
```

```
AISTUDIO_VECTORSTORE=qdrant PYTHONPATH=src python3 -m local_llm_bot.app.ingest \
  --corpus <name> --root data/corpora/<name>/uploads --force
```

***What happens when I delete an entire corpus — is it gone forever?*** No. The corpus folder moves to `~/.Trash/`. The folder is renamed to avoid conflicts with previous deletions — it will be named `AIStudio_<name>` or `AIStudio_<name>_<timestamp>` if a folder with that name already exists in Trash.

***How do I recover a corpus I accidentally deleted?*** First find the folder in Trash — it may have a timestamp suffix:

```
ls ~/.Trash/ | grep AIStudio_<name>
```

Then move it back and re-ingest:

```
mv ~/.Trash/AIStudio_<name> ~/Developer/AIStudio/data/corpora/<name>
# Or if it has a timestamp suffix:
mv ~/.Trash/AIStudio_<name>_<timestamp> ~/Developer/AIStudio/data/corpora/<name>
# Then re-ingest
```

Then re-upload the files via the UI (Add button) to restore the corpus in Qdrant.

***How do I ingest the SEC 10-K corpus?***

The SEC 10-K corpus is a special case. AIStudio provides `ais_download_sec_10k` because the SEC EDGAR filing system uses a specific access protocol — this automates what would otherwise be a complex multi-step download. But the ingest step that follows is **identical to ingesting any corpus you bring yourself**. The download step is what's special.

This is one of the few places in AIStudio where you'll run terminal commands directly. We expose it deliberately — understanding how large corpora are built is part of what makes AIStudio a learning tool, not just a product.

**Step 1 — Download the filings to ~/Downloads/sec_10k/ (~5 min, ~2 GB):**

```
ais_download_sec_10k
```

**Step 2 — Ingest using the AIStudio UI:**

Open AIStudio, create a new corpus named `sec_10k`, then upload the files from `~/Downloads/sec_10k/` using the **Upload** button. This is the same process as ingesting any corpus you build yourself — the download step is what's special.

Allow ~34 minutes for ingestion to complete.

> ***Why ~/Downloads?*** *You own the downloaded files — ~2 GB of SEC filings you may want to reuse, inspect, or build a different subset from. They stay in* `~/Downloads/sec_10k/` *until you decide what to do with them.*

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## Installing and Managing LLMs

#### *How do I see what models are installed?*

```
ollama list
```

#### *How do I install a new LLM?*

```
ollama pull llama3.1:8b       # ~5GB — recommended default, fast
ollama pull llama3.1:70b      # ~42GB — best quality, requires ~64GB RAM
ollama pull mistral:7b        # ~4.4GB — alternative option
```

Once pulled, the model appears automatically in the **Model** dropdown in the UI. No restart required.

***Which model should I choose?*** On Apple Silicon, `llama3.1:70b` and `llama3.1:8b` have identical query latency (~6–7s) once warm. Choose based on available RAM: - `llama3.1:8b` — 8GB RAM minimum, recommended for most users - `llama3.1:70b` — 64GB+ RAM, best answer quality - `mistral:7b` — good alternative on constrained hardware

#### *How do I remove a model I no longer need?*

```
ollama rm mistral:7b
```

See the Ollama model library for all available models.

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## Query Settings

The **Query Settings** panel in the sidebar controls how AIStudio retrieves and generates answers. These settings apply to every query until you change them.

**Top K** — *How many document chunks to retrieve* The number of passages retrieved from your corpus before generating an answer. Higher values give the model more context but increase latency slightly. - Default: `5` — good for most queries - Try `10` for complex questions spanning multiple documents - Try `3` for faster responses on focused questions

**Choosing Top K by query type:**

| Query type | Recommended Top K | Example |
|---|---|---|
| Point-in-time, single firm | 5 | "What was Goldman's CET1 ratio in 2024?" |
| Trend across multiple years | 10–15 | "How has JPMorgan's capital ratio changed 2022–2025?" |
| Cross-firm comparison | 10 | "Compare risk disclosures at JPMorgan, Goldman, and Citi" |
| General question | 5 | "What is the company's AI strategy?" |

If an answer seems incomplete or missing years/firms you expected — increase Top K first before rephrasing the question.

Adjust using the **Top K** input field in the sidebar.

**Temperature** — *How creative vs. precise the answer is* Controls randomness in the model's output. - `0.1–0.3` — precise, factual, stays close to the source documents. Best for financial data, specific numbers, compliance questions. - `0.5–0.7` — more varied, synthesizes across sources. Better for summaries. - Default: `0.3` — recommended starting point Adjust using the **Temperature** input field in the sidebar.

**Retrieval Mix** — *Balance between literal and conceptual search* Controls how the system finds relevant passages in your corpus.

- **Literal** (slider left, value 1.0) — prioritizes exact word and phrase matching (BM25). Use when you know the specific term, entity name, or abbreviation that appears in your documents. Example: searching for "CET1" by exact term.
- **Conceptual** (slider right, value 0.0) — prioritizes semantic meaning. Finds passages related to your query even when the document uses different phrasing. Example: asking about "capital strength" returns passages about CET1, Tier 1 capital, and capital ratios.
- **Default 0.5** — blends both. Works well for most queries.

**Choosing by query type:**

| Query type | Recommended setting | Why |
|---|---|---|
| Named entity, ticker, exact term | Literal (0.7–1.0) | BM25 matches the token directly |

| Query type | Recommended setting | Why |
|---|---|---|
| Thematic or conceptual question | Conceptual (0.0–0.3) | Semantic search handles paraphrase |
| Multi-firm comparison | Center (0.4–0.6) | Entity names + topic meaning both matter |
| Single firm, multiple years | Conceptual (0.0–0.2) | Year context is in documents; meaning guides retrieval |

Adjust using the **Retrieval Mix** slider in the Query Settings sidebar.

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## Understanding Citations

Every answer includes numbered citations — `[1]`, `[2]`, `[3]` — in the text. These refer to the source passages listed in the **References** panel on the right side of the screen.

**What citations tell you:** - Which document the information came from - Which page in that document

**What to do when an answer seems wrong:** Check the citations first. If the referenced chunks don't actually contain the claim — the model may be reasoning beyond its sources. Try increasing Top K, adjusting the Retrieval Mix toward Literal, or rephrasing to be more specific.

**Uncited claims:** if a sentence has no citation marker, the model is drawing on general context rather than a specific retrieved passage. Treat uncited claims with more caution, especially for specific numbers or facts.

**Opening the source:** click the citation chip in the References panel to see the full chunk text and open the source document.

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## Benchmark & Corpus Testing

The benchmark harness serves two purposes: - **Performance** — measures query latency per question - **Veracity** — validates answer quality against a questions file

***How do I run a benchmark on the demo corpus?***

```
ais_bench
```

***How do I benchmark a different corpus?***

```
ais_bench --corpus help
ais_bench --corpus sec_10k --top-k 10
```

#### *How do I test with a specific model or settings?*

```
ais_bench --corpus demo --model llama3.1:70b --top-k 10 --temperature 0.1
```

***How do I write my own test questions for a corpus?*** Create `benchmarks/<corpus_name>/<corpus_name>_questions.yaml`:

```
- topic: Getting Started
  questions:
    - id: start_aistudio
      question: How do I start AIStudio?
      notes: Should reference ais_start command
```

Run `ais_bench --corpus <n>` — the questions file is auto-detected by corpus name.

***Where do reports go?*** `benchmarks/<corpus>/reports/` — timestamped `.md` and `.json` pairs. The `.md` shows pass/fail per question with latency and citations.

For full benchmark documentation see HARNESS.pdf.

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## Troubleshooting

#### *UI shows "Error loading corpora" or "Ollama not running" on startup*

The browser opened before the backend finished starting. Hard-refresh ( `Cmd+Shift+R` ). If that doesn't fix it, check the terminal for a Qdrant WAL error (see below).

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#### *Ollama version mismatch warning — "client version is X.X.X"*

After a Homebrew update, Ollama's CLI binary may update while the background service still runs the old version. You'll see:

```
ollama version is 0.18.0
Warning: client version is 0.22.0
```

Fix:

```
brew services restart ollama
ais_stop
ais_start
```

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### *Qdrant WAL lock — "Can't init WAL: Resource temporarily unavailable"*

**What it is:** A corpus collection's write-ahead log was left locked from an unclean shutdown — force-quit terminal, power loss, or crash mid-write. Qdrant panics on the affected collection at startup. Other collections are unaffected.

**How you'll know:** Terminal shows the WAL error on `ais_start`, naming the specific collection. UI shows "Error loading corpora." Ingestion immediately fails.

**Fix:**

```
ais_stop
# Delete only the collection named in the panic message — e.g. aistudio_help
rm -rf ~/qdrant_storage/collections/aistudio_help
ais_start
```

Then re-ingest the corpus via the UI (Add button).

⚠️ Delete only the specific collection named in the error — not the entire `~/qdrant_storage/` folder. That would destroy all your corpora.

**Prevention:** Always stop with `ais_stop`. Never force-quit or close the terminal while AIStudio is running.

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### *Ingestion summary shows wrong file or chunk count*

Delete the corpus via the UI and re-ingest. The Qdrant collection may be out of sync with the uploads folder.

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***How do I rename a corpus?*** Use the **Rename** button in the corpus header in the UI. AIStudio renames the directory, updates the corpus metadata, and triggers a background re-index. You'll see a progress estimate for large corpora. Do not rename corpus folders manually on disk — use the UI only.

***How do I rename a file inside a corpus?*** AIStudio records file names at ingest time. Renaming a file after ingestion breaks citation lookup — queries will still return results but the "Open ↗" link will fail to resolve.

**Safe approach:** delete the file from the corpus using the **Delete** button in the Inspect view, then re-upload it with the correct filename via **Add**. AIStudio will re-ingest only the changed file.

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For architecture context, see docs/architecture_decisions.pdf. For getting started, see QUICKSTART.md.